

Mirror, Mirror on the Wall, Is AI Transforming Us All?

A Multi-Label Classification Pipeline for Analyzing AI's Behavioral Impact in Media, LLM Outputs, and Financial Markets

Synaptic Sparks

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Data + Traditional NLP

Deep Learning + LLM

Quantitative Finance

1

Environment & EDA

10,500 rows · 12 behavioral labels · avg 1.54 labels/sample · multi-label structure confirmed

2

Text Preprocessing

Lowercase · URL/HTML removal · whitespace normalization · 0 invalid rows dropped

3

Multi-Label Encoding

Multi-hot vectors · Multi label Stratified Shuffle Split · label sparsity not a limiting factor

4

TF-IDF + LR Baseline

One Vs Rest classifier · class-weighted · 2000 max iter · 2,000 TF-IDF features

0.9430

Micro-F1

0.9331

Macro-F1

0.9425

Weighted-F1

0.9216

Samples-F1

Top Label F1 Scores

Work, Jobs & Economy

0.982

Learning, Knowledge & Education

0.971

Technology & Interaction

0.962

Social Interaction & Relationships

0.959

Routine, Lifestyle & Behavior

0.957

Human Roles

0.948

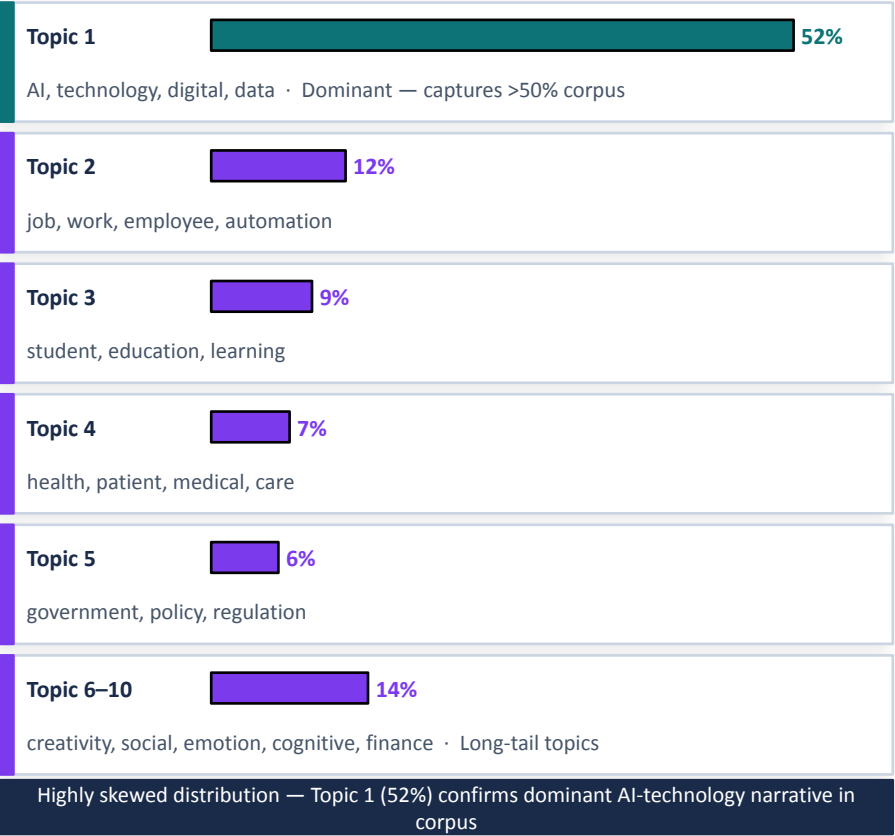
TF-IDF Coefficient Analysis

Label	Positive β (top features)	Negative β (suppressors)
Work & Economy	work, job, industry	education, gov
Learning & Edu	study, education, univ	industry, innovation
Tech & Interaction	technology, digital, chatbot	work, job
Society & Ethics	government, policy, law	job, study
Cognitive & Decision	decision, cognitive, bias	education, tech

Error Analysis Highlights

Mutual suppression: "job" promotes Work but suppresses Technology label
False Negative: "Employment Law" misses Work (legal-domain framing)
False Positive: "No more teachers..." incorrectly triggers Education
Weakest label: Cognitive (F1=0.848) — semantic overlap with Education

NMF Topic Modeling — 10 Latent Topics



Model Performance Comparison

Metric	TF-IDF + LR	DistilBERT	Δ (pp)
Micro-F1	0.9430	0.9214	+2.16
Macro-F1	0.9331	0.8862	+4.69 ★
Weighted-F1	0.9425	0.9149	+2.76
Samples-F1	0.9216	0.9115	+1.01

Why Does TF-IDF + LR Outperform DistilBERT?

Short text (10–15 tokens)

Headlines lack context; BoW features remain highly discriminative without long-range dependencies

Explicit lexical signals

Journalists use consistent domain vocabulary — strong TF-IDF signatures per behavioral category

Macro-F1 gap (+4.69 pp)

DistilBERT underperforms on rare labels with limited fine-tuning examples (< 200 samples)

Practical advantage

Minutes vs hours of training; no GPU required; full interpretability via LR coefficients

DistilBERT Architecture Details

Base model	distilbert-base-uncased
Fine-tuning epochs	3
Optimizer	AdamW
Scheduler	Linear warmup
Max token length	128 tokens
Classification head	Sigmoid (multi-label)
Loss function	BCEWithLogitsLoss
Device	CUDA / CPU fallback

Key Insight

For short, domain-specific text classification with clear lexical patterns, transformer models may not outperform well-tuned sparse baselines — especially with limited per-label training data.

74.21

χ^2 Statistic ($p < 10^{-7}$)

0.065

Cramér's V (effect size)

3 LLMs

Mistral / Qwen / Llama

12 labels

Behavioral categories

Label Distribution by LLM (%) — Top Categories

Category	Mistral	Qwen	Llama	$\Delta_{\max}$
Cognitive & Decision	23.2%	20.6%	20.2%	3.0pp
Routine & Lifestyle	15.5%	16.5%	16.0%	1.0pp
Creativity & Identity	8.4%	7.5%	7.2%	1.2pp
Sentiment	7.8%	8.6%	8.0%	0.8pp
Society & Ethics	7.0%	7.8%	8.6%	1.6pp

Prompt effects: "Teacher" → 45.7% Education · "Research" ≈ "Assistant" (identical) · Cognitive top-2 in ALL prompt types (21–36%)

LLM Specialization Profiles

Llama

Broadest scope

Leads 7/12 labels · Routine (0.384) · Sentiment (0.273) — most generalist behavioral range

Qwen

Most focused

Highest Cognitive (0.493) · Narrowest behavioral scope — concentrated analytical framing

Mistral

Most balanced

Leads Learning (0.273) · Health (0.252) · Social (0.224) — distributed across domains

Central Finding

All 3 LLMs converge on similar behavioral framings despite architectural differences. Cramér's V = 0.065 → statistically significant but practically small differences. This raises concerns about LLMs narrowing perspective diversity in AI discourse.

Analytical Pipeline

Configuration & Exposure Matrix

10 tickers (NVDA, AMD, META, GOOGL, MSFT, AMZN, SPY, QQQ...) ·
12-label industry exposure weight matrix

News Time Series Aggregation

Daily label counts from Dataset A · Weekly resampling for market data alignment

Market Data & GARCH(1,1)

yfinance price data · GARCH(1,1) via arch package · Annualized conditional volatility per ticker

Merge & Pearson Correlations

Weekly news-market panel · Pearson r between all 12 label proportions and vol/return cols

Granger Causality Test

Stats models granger causality tests · H0: label does not predict AI_vol_index · Max lag = 5 weeks

Industry Exposure Analysis

Label% x domain weight matrix · Enterprise leads (0.271) · Labor + Automation co-spike Aug 2025

Key Findings & Implications

Finding 1 — Predictive Causality

“Society, Ethics & Culture” Granger-causes AI sector volatility

- $p = 0.0007$
- lag = 5 weeks
- Sole significant predictor among 12 labels
- Forecasts shifts ~1 month ahead

Finding 2 — Asymmetric Volatility Effect

Ethics framing calms AI stocks, amplifies broad market

- The same ethics framing stabilizes AI stocks while amplifying broad market risk — a statistically significant asymmetry across asset classes

Finding 3 — EdAI & Emotion Signals

Education news boosts returns; emotion amplifies sector volatility

- Education coverage predicts positive returns; emotional coverage predicts higher volatility — rational vs. affective framing drives divergent investor behavior

Cross-domain conclusion: How media frames AI behavior is not just a linguistic signal — it is a quantifiable predictor of financial market dynamics.

GARCH(1,1) Annualized Volatility by Ticker

Ticker	Vol (%)	Risk Profile
AMD	48.4%	Pure-play AI hardware — highest volatility
NVDA	46.7%	AI chip demand cycles
META	36.0%	AI pivot + metaverse uncertainty
GOOGL	23.9%	Diversified revenue buffer
MSFT	21.4%	Enterprise moat, stable demand
SPY	17.8%	Broad market baseline

GRANGER CAUSALITY

p = 0.0007 · lag = 5 weeks

"Society, Ethics & Culture" is the sole significant predictor of AI sector volatility.
First empirical evidence linking behavioral news framing to financial markets.

Key Cross-Domain Correlations

-0.413 Society & Ethics → AMD vol

Ethics framing calms speculative AI stocks

+0.411 Society & Ethics → SPY vol

Same framing amplifies broad market uncertainty

+0.396 Society & Ethics → NVDA vol

EdAI news drives investor optimism

+ 0.314 Learning & Edu → AMZN returns

Rational framing reduces chip stock turbulence

+0.305 Emotion → AI volatility index

Emotional content amplifies sector swings

Methodology: yfinance data · GARCH(1,1) via arch package · Pearson correlations · Granger causality (statsmodels)

Thank you !

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